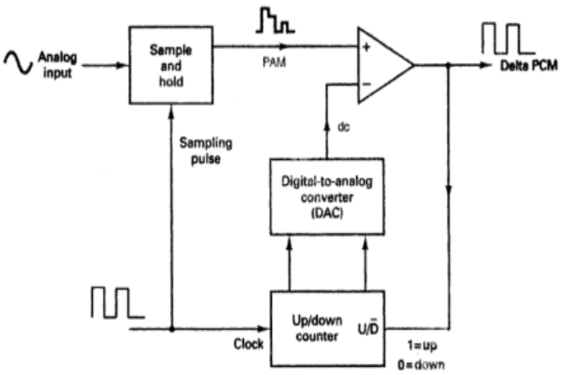
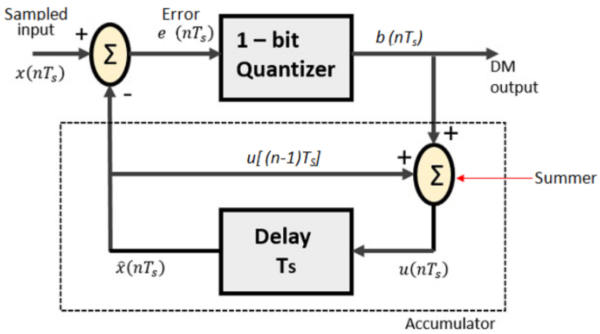
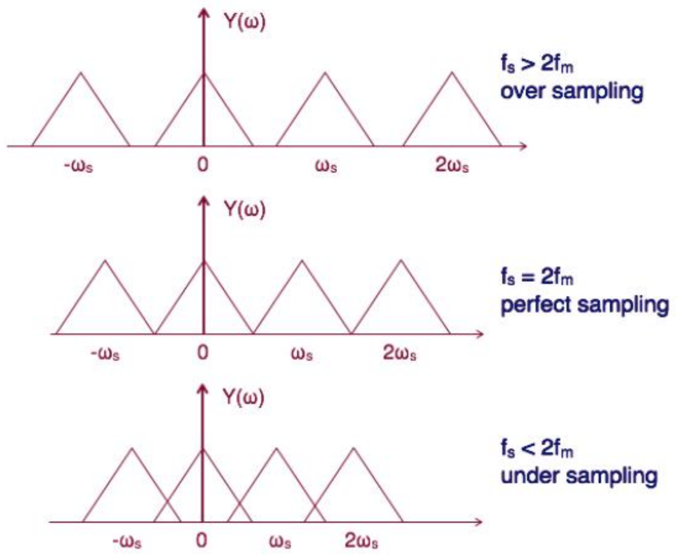
SOMAIYA VIDYAVIHAR UNIVERSITY

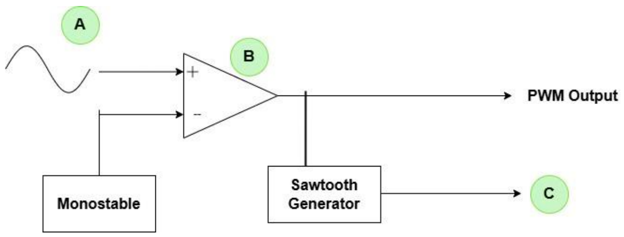
Maximum Marks: 50	**Semester: January 2025– April 2025** **Examination: End-Semester Examination**		**Duration: 2 Hrs.**
Programme code: 03 **Programme:** B.Tech Electronics and Telecommunication Engineering		**Class:** SY	**Semester:** IV **(SVU 2023)**
Institute/School/Department: K. J. Somaiya School of Engineering		**Name of the department:** EXTC	
Course Code: 216U03C403	**Name of the Course:** Communication Systems		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement (Marking Scheme)		Max. Marks
Q.1	Attempt any **two**		
i)	Suggest the method of transmitting analog signal digitally using only one bit. Draw neat labelled diagram for it and explain the process in brief.		05
	Name of Method: Delta modulation or Adaptive Delta Modulation	**1 Mark**	
	Diagram:  	**2 Marks**	
	Explanation	**2 marks**	
ii)	What should be the value of sampling frequency for reproducing analog signal back without any distortion? What will happen if this value not followed? Explain the cases with neat diagram.		05
	Sampling frequency should be greater than or equal to twice the maximum modulating signal frequency	**1 mark**	
	If sampling theorem is not followed there will be distortion in the reproduced signal	**1 mark**	



1 mark each

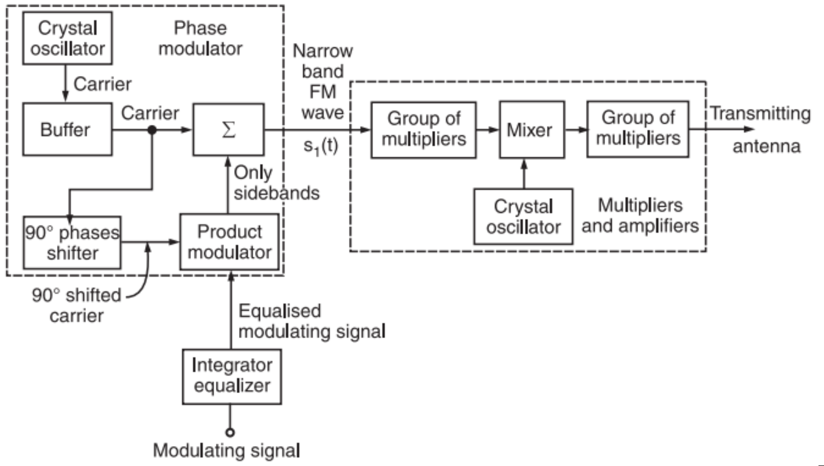
iii)	Identify the errors, if any in the following figure and redraw the figure with the appropriate labelling at point A, B and C. Mention one application of this method.	**05**



Input of inverting terminal – Sawtooth generator, Output- Monostable multivibrator	**1 mark**
Point A- Analog signal	**1 mark**
Point B- comparator	**1 mark**
Point C- PPM signal	**1 mark**
Application- remote controlled device, telecommunication, air traffic control etc.	**1 mark**

Q.2	Attempt any **one**	**10**
i)	Draw the labelled neat block diagram of generating frequency modulated signal from phase modulated signal. Explain the working in detail. Mention any one advantage of this method.	**10**

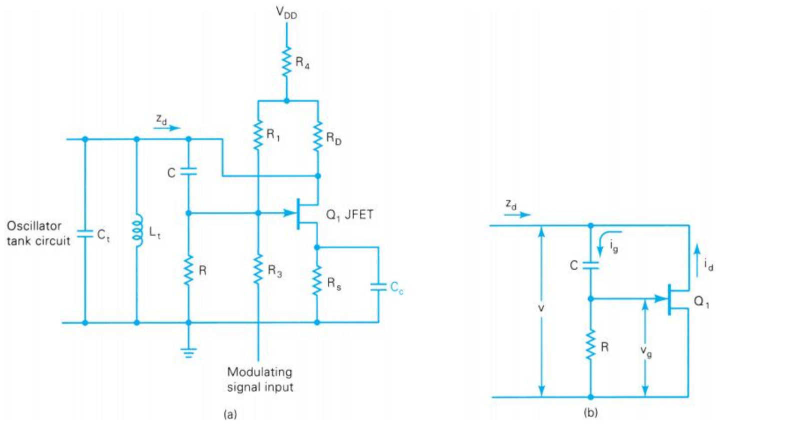
Neat and labelled block diagram of Armstrong method of FM generation	**5 marks**



Working	**4 Marks**
Advantage- high frequency deviations possible, more stable output	**1 mark**

ii) **A.** Explain the working of JFET based reactance FM modulator with neat diagram. **05**

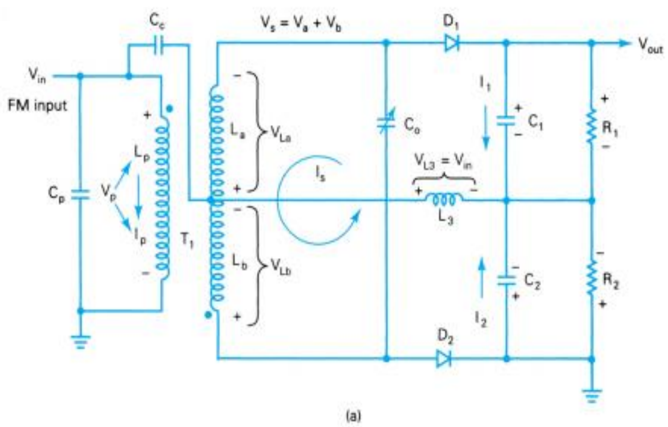
Neat labelled diagram	**3 marks**	**05**



Working	**2 marks**

B. Explain the process of demodulating FM signal using Foster-seeley discriminator with neat phasor diagrams.

Neat labelled circuit and phasor diagrams	**3 marks**



Working	**2 marks**

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Q.3	Attempt any **one**	**10**
i)	Draw basic block diagram to explain principle of SSB modulation. Why such modulation is advantageous? For SSB transmitter using filter method if, LF carrier frequency f_{LF} =190 KHz, MF carrier frequency f_{MF}=5 MHz, HF carrier f_{MF}=35 MHz, an audio input spectrum f_m =0 KHz to 3KHz. Sketch the frequency spectrum for the following points: (a) BPF1 out, BPF2 out, BPF3 out. (b) Balanced Modulator 1 out, Balanced Modulator 2 out, Balanced Modulator 3 out.	

Balanced Modulator 1 output **DSBSC: 187KHz, 190 KHz, 193 KHz**	**2 Marks**
Balanced Modulator 2 output **DSBSC: 4.807 MHz, 4.81 MHz, 5 MHz, 5.19 MHz, 5.193 MHz**	**2 Marks**
Balanced Modulator 3 output **DSBSC: 29.807 MHz, 29.81 MHz, 35 MHz, 40.19 MHz, 40.193 MHz**	**2 Marks**
BPF 1 output: 190KHz, 193 KHz **BPF 2 output: 5.19 MHz, 5.193 MHz** **BPF 3 output: 40.19 MHz, 40.193 MHz**	**4 Marks**

ii)	With reference to Amplitude modulation answer the following:	

- Mathematically prove that AM signal has carrier signal, upper and lower sideband signal.

Derivation of AM	5 Marks

Derivation of AM

Amplitude of Am Wave

$\rightarrow A = V_c + V_m$ (Read)

$$A = V_c + V_m \sin \omega_m t$$

$$A = V_c \left[1 + \frac{V_m}{V_c} \sin \omega_m t\right]$$

$m = \frac{V_m}{V_c}$ = Modulation Index of Am wave

$$\boxed{A = V_c [1 + m \sin \omega_m t]}$$

Derivation of AM Instantaneous amplitude of AM wave $V_{am}(t) = A \sin \omega_c t$ $V_{am}(t) = V_c[1 + m \sin \omega_m t] \sin \omega_c t$ $V_{am}(t) = V_c \sin \omega_c t + m V_c \underbrace{\sin \omega_m t \sin \omega_c t}_{\sin A \cdot \sin B}$ Note: $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$ ($A = \omega_c$, $B = \omega_m$) **Derivation of AM** $V_{am}(t) = \underbrace{V_c \sin \omega_c t}_{\text{Carrier signal}} + \underbrace{\frac{mV_c}{2} \cos(\omega_c - \omega_m)t}_{\text{Lower Sideband (LSB)}} - \underbrace{\frac{mV_c}{2} \cos(\omega_c + \omega_m)t}_{\text{Upper Sideband (USB)}}$	

- Mention the conditions of amplitudes of the modulating signal and carrier signal for successful information transmission and information loss.

For successful transmission Vm<Vc **Information loss Vm>Vc**	**1 mark**

- How power is distributed in conventional AM signal? Mention formulas

▪ In any electrical circuit, the power dissipated is equal to the voltage squared (rms) divided by the resistance. ▪ Mathematically power in unmodulated carrier is $P_c = \frac{(V_c/\sqrt{2})^2}{R} = \frac{V_c^2}{2R}$ P_c = *carrier power (watts)* V_c = *peak carrier voltage (volts)* R = *load resistance i.e antenna (ohms)*	1 mark

	■ The upper and lower sideband powers will be $$P_{usb} = P_{lsb} = \frac{(mV_c/2)^2}{2R} = \frac{m^2V_c^2}{8R}$$ ■ Rearranging in terms of P_c, $$P_{usb} = P_{lsb} = \frac{m^2}{4}\left(\frac{V_c^2}{2R}\right) = \frac{m^2}{4}P_c$$	1 mark	
	• What is difference between high level and low level modulation?		
	High level: Modulation takes place at the final element of final stage where carrier and modulating signal are at high power	**1 mark**	
	Low level: Modulation takes placed before amplification of carrier and modulating signal	**1 mark**	
Q.4	Attempt the following		**10**
	A. Define noise figure and noise factor with its formula. Why these parameters are important for communication system?		
	$$NF = \frac{SNR_{in}}{SNR_{out}}$$ Where: • SNR_{in} is the signal-to-noise ratio at the input, • SNR_{out} is the signal-to-noise ratio at the output. In decibels, this can be expressed as: $$NF_{dB} = 10\log_{10}(NF)$$ **Definition along with importance**	**2 marks**	
	B. Justify the name correlated and uncorrelated noise. Give one example of each.		

Correlated Noise

1 mark

- Intermodulation distortion is the generation of unwanted sum and difference frequencies produced when two or more signals mix in a nonlinear device.
  - The sum and difference freq are called cross product i.e mathematically

    Cross product = $mf_1 \pm nf_2$

    where f_1, f_2 = fundamental frequencies, $f_1 > f_2$

    m,n = positive integers
  - Unwanted cross-product freq can interfere with the info signals in a cct or with the info signal in other cct.

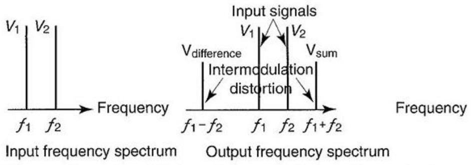


Figure: Correlated Noise (Intermodulation Distortion)

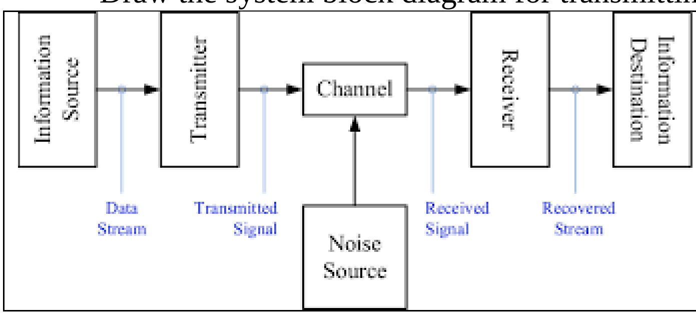
Uncorrelated Noise

1 mark

- Subdivided into 2 categories:
  - **External** Noise
    - Present in a received radio signal that has been introduced in the transmitting medium.
    - Source - atmospheric, extraterrestrial and man-made
  - **Internal** Noise
    - Introduced by the receiver itself.
    - Electrical interference generated within a device i.e create from the communication equipment.
    - Type - shot, transit time and thermal.

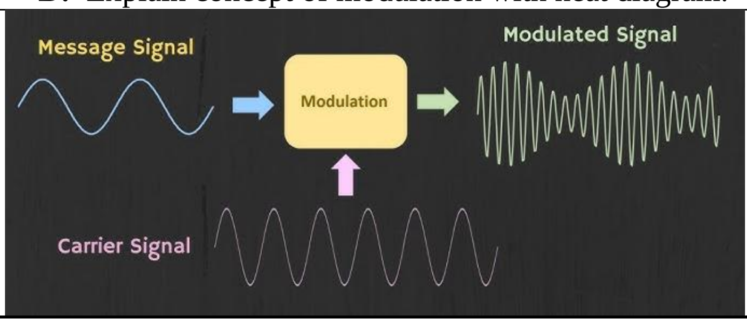
C. Jack has certain analog information to be sent to Jill at destination. Draw the system block diagram for transmitting this information.

2 marks

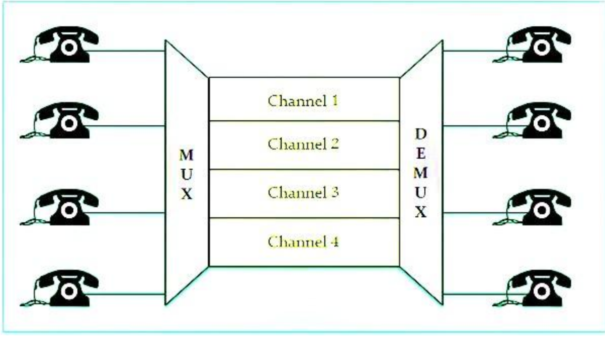


D. Explain concept of modulation with neat diagram.

2 marks



E. In the figure below if each channel occupies bandwidth of 105 KHz and guard band of 9 KHz is required to prevent interference between them, what is minimum bandwidth required for the link?



Bandwidth required is 105 KHz + 9 KHz + 105 KHz + 9 KHz + 105 KHz + 9 KHz + 105 KHz = 447 KHz	**2 marks**

Q.5	Attempt the following	10

A. For an AM broadcast band super heterodyne radio receiver with IF, RF and LO frequencies of 455 KHz, 600 KHz and 1055 KHz respective, determine (i) Image frequency (ii) IFRR for pre-selector of Q=100.

$f_{image} = f_{RF} + 2 \times IF = 1510\ KHz$	**1 mark**
$\rho = \frac{f_{image}}{f_{RF}} - \frac{f_{RF}}{f_{image}} = 2.1193$	
$IFRR = \sqrt{1 + \rho^2 + Q^2} = 211.94$	**1 marks**

B. Define the term sensitivity and selectivity for radio receiver.

• Sensitivity ▪ Minimum input signal (voltage) required to produce specified output. ▪ Noise floor • Input noise. • Selectivity ▪ Extent to which receiver capable of differentiating between desired signal and other frequencies.	**1 mark each**

C. In receiver ________ AGC waits for the signal to reach certain signal strength before applying gain adjustments, while ________AGC adjusts gain immediately based on the current signal strength.

Delayed	**1 mark**
Forward or simple	**1 mark**

D. Name the following figure A and B. What is major difference between them?

Fig. A

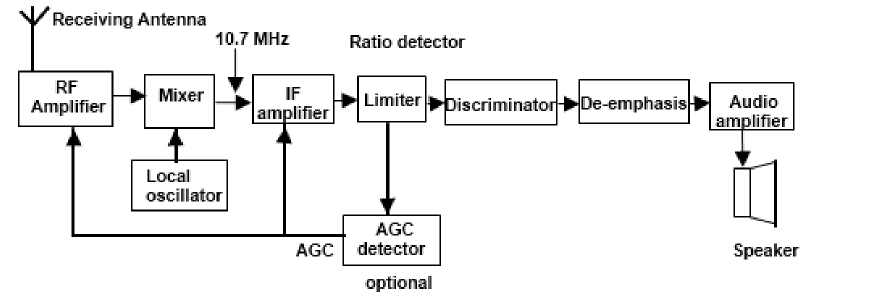


Fig. B

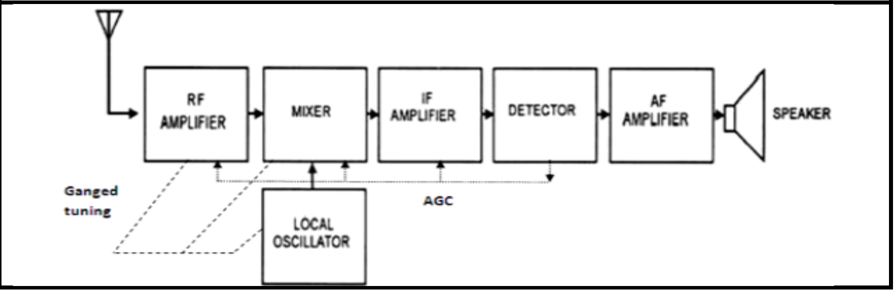


Figure A- FM receiver	**1 mark**
Figure B- AM receiver	**1 mark**

E. An FM wave is given by, $s(t) = 20\cos(8\pi \times 10^6 t + 9\sin(2\pi \times 10^3 t))$. Calculate the frequency deviation, highest and lowest frequencies attained by modulated signal.

$f_c = \frac{8\pi \times 10^6}{2\pi} = 4\ MHz$ $f_m = \frac{2\pi \times 10^3}{2\pi} = 1\ KHz$ $m_f = 9$	**1 mark**
$\Delta f = m_f \times f_m = 9000\ Hz$	
$f_{max} = f_c + \Delta f = 4009000\ Hz$	**1 mark**
$f_{max} = f_c - \Delta f = 3991000\ Hz$	